K G HARISH PATEL

Software Engineer | Python · JavaScript · FastAPI · REST APIs · SQL · Docker · LLMs · Computer Vision
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PROFESSIONAL SUMMARY

Final-year Computer Science engineer who builds and ships software end to end, from web features, REST APIs and SQL data models to computer vision and LLM applications. At ProEmbSys Technologies, I own a real-time PCB inspection service, turning ambiguous requirements from production and quality teams into tested, reliable releases. I shipped 7 features in a full-stack JavaScript and FastAPI application, wrapped a GraphRAG pipeline in a FastAPI microservice and fine-tuned open-source LLMs. I write clean, testable code, debug through root cause analysis and learn new systems quickly.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, C, HTML/CSS
Backend & APIs: FastAPI, RESTful API design, client-server architecture, OpenAPI, SQLAlchemy ORM, JWT authentication, role-based access control, JSON contracts, microservices fundamentals
Databases & Data: PostgreSQL, SQLite, relational schema design, data modeling, SQL query optimization, ACID transactions, data pipelines, data validation, data cleansing and transformation, Pandas, NumPy, analytics dashboards
AI & Vision: LLMs, RAG, GraphRAG, LoRA/QLoRA, PyTorch, Scikit-learn, OpenCV, YOLOv8, MobileNet-SSD, TensorFlow Lite, Structure-from-Motion, AirSim, Unreal Engine, NetworkX, agentic AI tools
Tools & Practices: Git, GitHub Pages, Docker, Jenkins CI/CD, Linux, Agile/Scrum, data structures & algorithms, object-oriented design, design patterns, unit, integration & regression testing, TDD, test automation, debugging, root cause analysis, performance profiling, code review, requirement gathering, documentation

EXPERIENCE

Project Intern, Edge AI & Machine Vision | ProEmbSys Technologies Pvt. Ltd., Bengaluru **Sep 2026 – Present**
- Built a 0-to-1 real-time inspection service in Python and OpenCV that detects 4 PCB defect classes, owning design, implementation, testing and delivery.
- Translated ambiguous production and quality team requirements into testable acceptance criteria, validating every release on precision, recall and mAP@0.5.
- Debugged false positives through per-class root cause analysis, comparing YOLOv8 and MobileNet-SSD models and shipping only non-regressing changes.
- Profiled latency, FPS and memory footprint on resource-constrained hardware, documenting performance trade-offs that guided deployment decisions.

Artificial Intelligence Intern | Tech Mindsparc Innovations Pvt. Ltd., Remote **Sep 2024 – Dec 2024**
- Developed maintainable, reusable Python modules (Scikit-learn, Pandas, NumPy) for supervised classification, from data preparation to release.
- Engineered data-validation and feature-extraction pipelines that caught bad input early and improved model accuracy over the baseline.
- Implemented an automated, metric-based quality gate that blocked any regressing change, keeping every result reproducible and reviewable.
- Delivered iterative features over 4 months on a remote Agile team, reporting in stand-ups and applying code review feedback each sprint.
- Researched and compared classification approaches, presenting data-backed insights that informed model selection.

PROJECTS

LibCheck – Full-Stack Library Management System | *JavaScript, FastAPI, REST APIs, SQLAlchemy, SQLite* **Dec 2025**
- Built the JavaScript UI of a 3-tier app and integrated it with a FastAPI REST API across 4 OpenAPI-documented modules.
- Shipped 7 user-facing features (smart search, analytics dashboards, email notifications) with unit and integration tests.
- Secured the REST API with JWT authentication and role-based access control, adding audit logging for every state change.
- Modeled a normalized SQLAlchemy schema with transactions that keep loan, fine and availability updates consistent.

GraphRAG Knowledge Service | *Python, FastAPI, LLMs, NetworkX, Docker* **Aug 2026**
- Architected an LLM pipeline extracting entities and relations from unstructured documents into a NetworkX knowledge graph.
- Exposed retrieval and Q&A as a FastAPI microservice with defined JSON request/response contracts for client integration.
- Benchmarked answer accuracy against a chunk-based RAG baseline, validating LLM outputs against defined success criteria.
- Containerized the service with Docker for reproducible deployments across development and demo environments.

LoRA / QLoRA LLM Fine-Tuning | *Python, PyTorch, LoRA, QLoRA* **Jul 2026**
- Fine-tuned an open-source LLM with parameter-efficient LoRA and QLoRA adapters for a task-specific use case.
- Profiled GPU memory and compute usage to document LoRA-versus-QLoRA performance and resource trade-offs.
- Automated dataset validation checks for format, duplicates and quality that ran before every training job.
- Measured each change against a pre-fine-tuning baseline on defined evaluation metrics to isolate its effect.

Drone Navigation & 3D Reconstruction | *Python, AirSim, Unreal Engine, OpenCV* **Jun 2026**
- Developed a controller-based drone navigation simulation in AirSim and Unreal Engine to generate repeatable flight data.
- Automated a synchronized OpenCV data-acquisition pipeline pairing every RGB frame with its camera pose.
- Reconstructed 3D scenes from the captured frames using Structure-from-Motion (SfM).
- Placed 2nd at the Innovotsava 2026 Drone Fair, judged on innovation and technical problem-solving.

EDUCATION

B.E. in Computer Science and Engineering | Maharaja Institute of Technology, Mysore **Sep 2023 – Aug 2027 (Expected)**
Relevant Coursework: DSA, OOP, DBMS (SQL), Operating Systems, Computer Architecture, Machine Learning

LEADERSHIP & CERTIFICATIONS

- Co-Founder & Content Lead, Coding Kannadiga (YouTube), Jul 2025 – Present: published 90 tutorials on Python, DSA and AI.
- JPMorgan Chase Software Engineering Job Simulation (Forage); Deloitte Cyber Job Simulation (Forage); AI Aware (Intel, 2025).